

# A Reduced Newtonian Model of Tidally Deformable Pseudo-Fluid Planets: Full Modal Formulation and Quadrupole Implementation

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**Abstract**—This paper develops a Newtonian model for tidally interacting planets whose surfaces deviate from exact sphericity through damped pseudo-fluid modes. The goal is not a full hydrodynamic interior simulation, but a structured reduced model that still contains the key mechanisms of tidal evolution: tidal bulges, gravitational backreaction of the bulges, spin torques, orbital circularization, spin synchronization, and obliquity damping. First, a general deformable-body formulation is written in which each planet has orbital motion, rigid-body orientation, spin, and an arbitrary set of surface modes. Second, a feasible all-mode approximation is introduced using body-frame symmetric trace-free tensors and generalized rheology with relaxation branches. Third, a quadrupole truncation is derived and written in a form directly suitable for numerical integration. Compared with the simplest quadrupole tide model, the upgraded version includes shape-dependent spin inertia, rotational flattening, and frequency-dependent lag through internal memory variables. The relation of the model to real planetary tide problems, and its status as a structured toy model rather than a full geophysical code, are discussed explicitly.

**Index Terms**—Tides, celestial mechanics, deformable bodies, spherical harmonics, quadrupole approximation, spin-orbit coupling, rheology.

## I. INTRODUCTION

Many problems in celestial mechanics lie between two standard idealizations. At one extreme, planets are treated as point masses and evolve only through Newtonian  $N$ -body gravity. At the other extreme, one solves the full equations of fluid or viscoelastic motion in self-gravitating rotating bodies, including stratification, realistic rheology, and thermal physics. The first approach is simple but cannot represent tides. The second is physically rich but expensive and often problem-specific.

The purpose of this work is to formulate a middle-ground model with the following properties:

- 1) full three-dimensional Newtonian orbital dynamics,
- 2) explicit deformation degrees of freedom,
- 3) deformation-dependent gravity,
- 4) spin and orientation dynamics,
- 5) dissipative tidal lag,
- 6) a systematic path from a general modal model to a compact implementation model.

The resulting framework is still a reduced model. It is much closer to a “structured toy model” than to a full planetary interior simulation. However, it is not an arbitrary toy. It

is designed so that the couplings between orbit, spin, and deformation have the correct geometric form, and so that increasingly refined approximations can be introduced without changing the overall architecture.

Compared with the simplest damped-quadrupole tide model, the present version includes three important improvements:

- the spin inertia depends on the tidal deformation,
- rotational flattening is included through the same shape-dependent spin inertia that appears in the reduced spin kinetic energy, so the rotational forcing and the energy bookkeeping use one common coefficient,
- the dissipative response can be modeled by one or more relaxation branches rather than a single constant time-lag term.

These additions make the reduced model substantially more faithful to real tidal systems while keeping the equations in ordinary differential equation (ODE) form.

## II. KINEMATICS AND VARIABLES

Consider  $N$  planets indexed by  $a = 1, \dots, N$ . For each body we introduce

- mass  $M_a$ ,
- reference radius  $R_a$ ,
- center-of-mass position  $X_a(t) \in \mathbb{R}^3$ ,
- orientation matrix  $Q_a(t) \in SO(3)$ ,
- body-frame angular velocity  $\Omega_a(t) \in \mathbb{R}^3$ .

The orientation evolves according to

$$\dot{Q}_a = Q_a \hat{\Omega}_a, \quad (1)$$

where  $\hat{\Omega}_a$  is the skew-symmetric matrix satisfying

$$\hat{\Omega}_a v = \Omega_a \times v. \quad (2)$$

A point  $y$  in the body frame corresponds to inertial position

$$x = X_a + Q_a y. \quad (3)$$

This explicit separation between inertial and body frames is essential. It avoids fictitious torques in the reduced equations and makes it possible to describe tides by body-frame variables while still computing forces and torques in the inertial frame.

### III. SURFACE DESCRIPTION AND EXACT DEFORMABLE BODY

#### A. Surface shape

Assume the body remains star-shaped with respect to its center. Its surface in the body frame is written as

$$r = \mathcal{R}_a(\hat{y}, t) = R_a(1 + u_a(\hat{y}, t)), \quad (4)$$

where  $\hat{y} \in S^2$  and  $u_a$  is the dimensionless shape field.

Expand the shape in spherical harmonics:

$$u_a(\hat{y}, t) = \sum_{\ell=0}^{\infty} \sum_{m=-\ell}^{\ell} q_{a\ell m}(t) Y_{\ell m}(\hat{y}). \quad (5)$$

The occupied body-frame volume is

$$V_a(t) = \{y \in \mathbb{R}^3 : 0 \leq |y| < \mathcal{R}_a(\hat{y}, t)\}. \quad (6)$$

#### B. Constraints

For an incompressible pseudo-fluid, the volume is fixed:

$$\frac{1}{3} \int_{S^2} \mathcal{R}_a(\hat{y}, t)^3 d\Omega = \frac{4\pi}{3} R_a^3. \quad (7)$$

The body-frame center of mass is fixed at the origin:

$$\int_{V_a(t)} y d^3y = 0. \quad (8)$$

At linear order, these conditions eliminate independent  $\ell = 0$  and  $\ell = 1$  modes. The tidal content therefore begins at  $\ell = 2$ .

#### C. Exact gravity

Assume uniform density inside the deformed body:

$$\rho_a(t) = \frac{M_a}{\text{Vol}(V_a(t))}. \quad (9)$$

The exact Newtonian potential of body  $a$  is

$$\Phi_a(x, t) = -G\rho_a \int_{V_a(t)} \frac{d^3y}{|x - X_a - Q_a y|}. \quad (10)$$

For  $x$  outside the body, this admits a multipole expansion. Let

$$s = Q_a^\top (x - X_a). \quad (11)$$

Then

$$\Phi_a(x, t) = -G \sum_{\ell=0}^{\infty} \sum_{m=-\ell}^{\ell} \frac{4\pi}{2\ell+1} \frac{Q_{a\ell m}(t)}{|s|^{\ell+1}} Y_{\ell m}(\hat{s}), \quad (12)$$

where

$$Q_{a\ell m}(t) = \int_{V_a(t)} \rho_a r^\ell Y_{\ell m}^*(\hat{y}) d^3y. \quad (13)$$

Equivalently,

$$Q_{a\ell m}(t) = \frac{\rho_a}{\ell+3} \int_{S^2} \mathcal{R}_a(\hat{y}, t)^{\ell+3} Y_{\ell m}^*(\hat{y}) d\Omega. \quad (14)$$

This is the exact sense in which the deformation changes the gravitational field.

#### D. Exact pseudo-fluid motion

Let the internal motion be irrotational in the body frame:

$$w_a(y, t) = \nabla_y \phi_a(y, t), \quad \nabla_y^2 \phi_a = 0 \quad \text{in } V_a(t). \quad (15)$$

The inertial velocity is then

$$v_a(x, t) = \dot{X}_a + Q_a(\Omega_a \times y + \nabla_y \phi_a). \quad (16)$$

If

$$F_a(y, t) = |y| - \mathcal{R}_a(\hat{y}, t), \quad (17)$$

the kinematic boundary condition is

$$\partial_t F_a + \nabla_y F_a \cdot \nabla_y \phi_a = 0 \quad \text{on } F_a = 0. \quad (18)$$

#### E. Exact energies

The orbital kinetic energy is

$$T_{\text{orb}} = \sum_a \frac{1}{2} M_a |\dot{X}_a|^2. \quad (19)$$

The internal-plus-spin kinetic energy is

$$T_{\text{int+spin}} = \sum_a \frac{\rho_a}{2} \int_{V_a(t)} |\Omega_a \times y + \nabla \phi_a|^2 d^3y. \quad (20)$$

The self-gravity is

$$U_a^{\text{self}} = -\frac{G\rho_a^2}{2} \int_{V_a(t)} \int_{V_a(t)} \frac{d^3y d^3z}{|y - z|}. \quad (21)$$

A material or restoring energy may be added as a functional of the shape:

$$U_a^{\text{mat}} = U_a^{\text{mat}}[u_a]. \quad (22)$$

The exact mutual interaction is

$$U_{ab} = -G\rho_a\rho_b \int_{V_a(t)} \int_{V_b(t)} \frac{d^3y d^3z}{|X_a - X_b + Q_a y - Q_b z|}. \quad (23)$$

The exact Lagrangian is

$$L = T_{\text{orb}} + T_{\text{int+spin}} - \sum_a (U_a^{\text{self}} + U_a^{\text{mat}}) - \sum_{a < b} U_{ab}. \quad (24)$$

This exact moving-boundary model is conceptually useful, but it is not practical as a general-purpose tide integrator. The moving-domain Laplace problem, the nonlinear volume constraints, and the double volume integrals make it far more expensive than desired. The rest of the paper develops a feasible approximation that keeps the same orbit-spin-shape structure.

### IV. FEASIBLE FULL MODAL MODEL

#### A. STF mode variables

For actual computation, it is more convenient to replace the spherical harmonic amplitudes by body-frame symmetric trace-free (STF) tensors. For each degree  $\ell \geq 2$ , let

$$A_a^{(L)}(t) \quad (25)$$

be a body-frame STF tensor of rank  $\ell$ . This is equivalent to the set of coefficients  $\{q_{a\ell m}\}_{m=-\ell}^{\ell}$ .

The corresponding mass multipole is written as

$$M_a^{(L)} = \kappa_{a\ell} A_a^{(L)}. \quad (26)$$

For later use, recall that for any rank-2 tensor  $B$  its symmetric trace-free part is

$$\text{STF}(B) = \frac{1}{2}(B + B^\top) - \frac{1}{3}(\text{tr } B)I_3. \quad (27)$$

In particular,

$$\text{STF}(\Omega_a \Omega_a^\top) = \Omega_a \Omega_a^\top - \frac{1}{3}|\Omega_a|^2 I_3. \quad (28)$$

### B. Corotational derivative

Because the modes are stored in the rotating body frame, the natural time derivative is the corotational derivative

$$(D_t A)_{i_1 \dots i_\ell} = \dot{A}_{i_1 \dots i_\ell} + \sum_{k=1}^{\ell} \epsilon_{i_k p q} \Omega_{a,p} A_{i_1 \dots q \dots i_\ell}. \quad (29)$$

Its second derivative is

$$D_t^2 A = D_t(D_t A). \quad (30)$$

This is the correct body-frame transport law for a tensorial tidal mode.

### C. External tidal moments

Let  $\Phi_{\text{ext},a}$  be the potential generated by all bodies other than  $a$ . The inertial STF tidal moments at  $X_a$  are

$$E_{a,\text{in}}^{(L)} = \left[ \nabla^{(L)} \Phi_{\text{ext},a}(x) \right]_{x=X_a}. \quad (31)$$

The body-frame moments are

$$E_{a,\text{body}}^{(L)} = Q_a^\top \dots Q_a^\top E_{a,\text{in}}^{(L)}. \quad (32)$$

### D. Shape-dependent spin inertia

A key improvement over the simplest tide models is that the spin inertia is allowed to depend on the deformation. Denote the body-frame spin inertia by

$$J_a = J_a[A]. \quad (33)$$

For small deformations, this may be expanded linearly in the retained modes. In particular, in the quadrupole sector one will use

$$J_a(S_a) = J_{a0} I_3 - c_{J,a} S_a + O(S_a^2). \quad (34)$$

This correction is important for leading-order consistency between tidal torques and spin dynamics. In the reduced quadrupole closure developed below, the same shape dependence is also used to represent the retained rotational-flattening response, rather than introducing a second independent conservative storage term at the same order.

### E. Generalized rheology

A single linear damping term corresponds essentially to a constant time-lag response. To allow more flexible frequency dependence while keeping the system in ODE form, introduce memory variables

$$Z_{a\ell n}^{(L)}(t), \quad n = 1, \dots, N_{r,a\ell}, \quad (35)$$

where  $n$  indexes relaxation branches.

Each branch has stiffness  $c_{a\ell n} \geq 0$  and relaxation time  $\tau_{a\ell n} > 0$ .

### F. Reduced full-model Lagrangian

A convenient all-mode reduced Lagrangian is

$$L = \sum_a \left[ \frac{1}{2} M_a |\dot{X}_a|^2 + \frac{1}{2} \Omega_a^\top J_a[A] \Omega_a + \sum_{\ell \geq 2} \frac{\mu_{a\ell}}{2} \|D_t A_a^{(L)}\|^2 - \sum_{\ell \geq 2} \frac{\mu_{a\ell} \omega_{a\ell}^2}{2} \|A_a^{(L)}\|^2 \right] - U(X, Q, A), \quad (36)$$

where  $\mu_{a\ell}$  is a modal inertia and  $\omega_{a\ell}$  is the reference mode frequency.

The interaction energy is approximated by

$$U = - \sum_{a < b} \frac{G M_a M_b}{|X_a - X_b|} + U_{\text{tide}} + U_{\text{rot}}, \quad (37)$$

with

$$U_{\text{tide}} = - \sum_a \sum_{\ell \geq 2} \frac{1}{\ell!} M_{a,\text{in}}^{(L)} E_{a,\text{in}}^{(L)}, \quad (38)$$

and

$$U_{\text{rot}} = \sum_a \beta_{a2} A_a^{(ij)} \text{STF}(\Omega_{ai} \Omega_{aj}), \quad (39)$$

where only the  $\ell = 2$  sector contributes. At the full modal level this term is the conservative rotational coupling whose gradient with respect to the quadrupole mode produces the centrifugal forcing retained in the shape equation.

### G. All-mode evolution equations

The reduced full-model equations are

$$M_a \ddot{X}_a = -\nabla_{X_a} U, \quad (40)$$

$$\dot{Q}_a = Q_a \hat{\Omega}_a, \quad (41)$$

and

$$\dot{\Pi}_a + \Omega_a \times \Pi_a = \tau_a, \quad \Pi_a = \frac{\partial L}{\partial \Omega_a}, \quad (42)$$

where  $\Pi_a$  is the total body-frame angular momentum retained by the model.

For the shape modes, use

$$\begin{aligned} \mu_{a\ell} D_t^2 A_a^{(L)} + \mu_{a\ell} \omega_{a\ell}^2 A_a^{(L)} + \sum_{n=1}^{N_{r,a\ell}} c_{a\ell n} (A_a^{(L)} - Z_{a\ell n}^{(L)}) \\ = -\kappa_{a\ell} E_{a,\text{body}}^{(L)} - \delta_{\ell 2} \beta_{a2} \text{STF}(\Omega_a \Omega_a^\top)^{(L)}. \end{aligned} \quad (43)$$

The relaxation branches satisfy

$$\tau_{a\ell n} D_t Z_{a\ell n}^{(L)} = A_a^{(L)} - Z_{a\ell n}^{(L)}. \quad (44)$$

Equation (43) contains three distinct sources of deformation: the body's own restoring tendency, the external tide, and the centrifugal flattening term in the  $\ell = 2$  sector. In the rank-2 case, the rotational forcing is exactly the STF tensor defined in (28), which removes the isotropic part and retains only the pure quadrupolar flattening. In the full modal model this rotational term may be viewed as arising from the conservative coupling  $U_{\text{rot}}$ . In the reduced quadrupole closure used later, the same effect is folded into the shape-dependent inertia coefficient so that the reduced spin equation and the reduced energy remain consistent.

#### H. Comments on angular momentum

The reduced model is most consistent when the spin equation is derived from the same Lagrangian that defines the mode kinetic energy. In that case  $\Pi_a$  includes not only the rigid-spin part  $J_a[A]\Omega_a$ , but also the angular momentum carried by the corotational mode kinetic terms. For a strictly leading-order quadrupole implementation, one may keep only the dominant shape dependence of the rigid-spin inertia and neglect higher-order modal contributions to  $\Pi_a$ . This is the approximation most often taken in practice. The resulting quadrupole closure therefore preserves the leading spin–shape coupling while sacrificing exact global angular-momentum conservation at higher order, because the angular momentum stored in the deformation itself is not evolved explicitly.

### V. PHYSICAL INTERPRETATION OF THE FULL MODAL MODEL

The full modal model contains the main qualitative tidal effects.

#### A. Tidal bulges

In the slow-forcing limit, the modes approach a quasi-static response proportional to the external tidal field:

$$A_a^{(L)} \approx -\frac{\kappa_{a\ell}}{\mu_{a\ell}\omega_{a\ell}^2} E_{a,\text{body}}^{(L)}, \quad (45)$$

up to corrections from rheology and rotation. This is the tidal bulge.

#### B. Rotational flattening

Even without an external perturber, the reduced energy contains the spin–shape coupling

$$U_{\text{rot},a} = \beta_{a2} A_a^{(ij)} \text{STF}(\Omega_{ai}\Omega_{aj}), \quad (46)$$

whose gradient generates the term

$$-\delta_{\ell 2} \beta_{a2} \text{STF}(\Omega_a \Omega_a^\top) \quad (47)$$

in (43). Thus an isolated rapidly rotating body does not relax to a perfect sphere; instead it approaches the rotationally flattened equilibrium selected by this conservative coupling, modified by any dissipation present.

#### C. Lag, synchronization, and circularization

The relaxation branches cause the deformation to lag behind the forcing. The lagged bulge exerts a torque on the spin and modifies the force on the orbit. As a result, the model supports:

- spin synchronization or pseudo-synchronization,
- obliquity damping,
- orbital circularization,
- exchange of angular momentum between orbit, spin, and tides.

### VI. QUADRUPOLE REDUCTION

The most useful implementation model keeps only the  $\ell = 2$  sector.

#### A. Quadrupole shape tensor

Represent the body-frame quadrupole deformation by a symmetric trace-free matrix

$$S_a(t) \in \mathbb{R}^{3 \times 3}, \quad S_a = S_a^\top, \quad \text{tr } S_a = 0. \quad (48)$$

The perturbed surface is

$$r = R_a(1 + \hat{n}^\top S_a \hat{n}). \quad (49)$$

#### B. Quadrupole moment and inertia

The inertial mass quadrupole is

$$I_a = c_{Q,a} Q_a S_a Q_a^\top. \quad (50)$$

For the normalization used in many reduced models,

$$c_{Q,a} = \frac{2}{5} M_a R_a^2. \quad (51)$$

The body-frame spin inertia is taken as

$$J_a(S_a) = J_{a0} I_3 - \tilde{c}_{J,a} S_a, \quad (52)$$

where  $J_{a0}$  is the spherical reference inertia and  $\tilde{c}_{J,a}$  is the effective reduced spin–shape coefficient. If one starts from a split into a direct inertia-response coefficient  $c_{J,a}$  and a separate conservative rotational-flattening coefficient  $\alpha_a$ , then the reduced quadrupole closure requires the combination

$$\tilde{c}_{J,a} = c_{J,a} + 2\alpha_a. \quad (53)$$

For a homogeneous reference body,

$$J_{a0} = \frac{2}{5} M_a R_a^2. \quad (54)$$

Equation (52) is a linearized leading-order correction. Because  $S_a$  is trace-free, this closure keeps  $\text{tr } J_a = 3J_{a0}$  fixed. That should be understood as a small-deformation linearization:  $O(S_a^2)$  corrections would in general modify the trace, so (52) is not exact for large deformation. Within the reduced quadrupole closure, however, (52) is the correct place to retain the leading rotational-flattening feedback, because the same coefficient must appear both in the spin kinetic energy and in the rotational forcing of the shape equation.

### C. External tidal tensor

Let

$$D_{ab} = X_b - X_a, \quad d_{ab} = |D_{ab}|, \quad \hat{D}_{ab} = D_{ab}/d_{ab}. \quad (55)$$

The inertial quadrupolar tidal tensor produced by the other bodies is

$$E_{a,\text{in}} = \sum_{b \neq a} \frac{GM_b}{d_{ab}^3} \left( I_3 - 3\hat{D}_{ab}\hat{D}_{ab}^\top \right). \quad (56)$$

The body-frame tidal tensor is

$$E_{a,\text{body}} = Q_a^\top E_{a,\text{in}} Q_a. \quad (57)$$

Because the quadrupole interaction energy is

$$U_{\text{tide},a}^{(2)} = -\frac{1}{2} I_a : E_{a,\text{in}} = -\frac{c_{Q,a}}{2} S_a : E_{a,\text{body}}, \quad (58)$$

the reciprocal tidal forcing that appears in the shape equation is

$$\lambda_a E_{a,\text{body}}, \quad \lambda_a \equiv \frac{c_{Q,a}}{2}. \quad (59)$$

This coefficient is required for consistency between the quadrupole force, torque, interaction energy, and the internal mode dynamics.

In the reduced quadrupole closure adopted here, rotational flattening is not kept as a separate stored term  $U_{\text{rot},a}^{(2)}$ . Instead it is absorbed into the same linearized inertia coefficient that appears in (52). Because  $S_a$  is symmetric and trace-free,

$$\frac{1}{2} \Omega_a^\top J_a(S_a) \Omega_a = \frac{1}{2} J_{a0} |\Omega_a|^2 - \frac{1}{2} \tilde{c}_{J,a} S_a : \text{STF}(\Omega_a \Omega_a^\top), \quad (60)$$

so the rotational contribution implied by the reduced spin kinetic energy is

$$-\frac{1}{2} \tilde{c}_{J,a} \text{STF}(\Omega_a \Omega_a^\top). \quad (61)$$

This is the coefficient that must appear in the reduced quadrupole shape equation. Adding a second independent term  $\alpha_a S_a : \text{STF}(\Omega_a \Omega_a^\top)$  to the reduced mechanical energy would double count the same conservative coupling unless one also kept the corresponding extra contribution to the reduced spin momentum.

### D. Corotational derivative

For any body-frame rank-2 tensor, define

$$D_t S_a = \dot{S}_a + [\hat{\Omega}_a, S_a], \quad (62)$$

where

$$[A, B] = AB - BA. \quad (63)$$

Then

$$D_t^2 S_a = D_t(D_t S_a). \quad (64)$$

It is convenient to introduce the auxiliary variable

$$W_a = D_t S_a. \quad (65)$$

### E. Quadrupole shape equation with rheology

The quadrupole shape equation is

$$D_t^2 S_a + 2\gamma_a D_t S_a + \omega_{a2}^2 S_a + \sum_{n=1}^{N_{r,a}} c_{an} (S_a - Z_{an}) = -\lambda_a E_{a,\text{body}} - \frac{1}{2} \tilde{c}_{J,a} \text{STF}(\Omega_a \Omega_a^\top). \quad (66)$$

Each relaxation branch obeys

$$\tau_{an} D_t Z_{an} = S_a - Z_{an}. \quad (67)$$

The coefficient  $\gamma_a$  represents direct viscous-like damping of the quadrupole rate, while the branch terms represent delayed internal response. A single branch is often enough for a useful toy model; several branches can approximate a broader frequency-dependent rheology.

### F. Spin equation

In the particular leading-order quadrupole closure used by the reference implementation, the body-frame spin equation is

$$\frac{d}{dt} (J_a(S_a) \Omega_a) + \Omega_a \times (J_a(S_a) \Omega_a) = 0. \quad (68)$$

Expanding the time derivative gives

$$J_a(S_a) \dot{\Omega}_a = -\dot{J}_a(S_a) \Omega_a - \Omega_a \times (J_a(S_a) \Omega_a). \quad (69)$$

Since  $W_a = D_t S_a$ , one also has

$$\dot{S}_a = W_a - [\hat{\Omega}_a, S_a], \quad (70)$$

so that in the linearized closure

$$\dot{J}_a(S_a) = -\tilde{c}_{J,a} \dot{S}_a. \quad (71)$$

This is the simplest leading-order correction that accounts for the fact that the tidal bulge changes the moment of inertia. In the present reduced model, it is paired with the deliberate omission of the deformation's own angular momentum from  $\Pi_a$ , as discussed above.

The same truncated quadrupole potential still defines a formal torque, but in this reduced closure that torque is not inserted as a second independent term on the right-hand side of (68). The orientational dependence of the interaction is already represented through the body-frame transport term in (70); adding the explicit torque again would double count the same conservative  $Q_a$ - $S_a$  coupling and produces a secular drift in the “mechanical plus dissipated” energy diagnostic of the implementation.

### G. Diagnostic quadrupole torque

The inertial quadrupole torque associated with the same truncated interaction energy is

$$N_{a,\text{in}} = 3 \sum_{b \neq a} \frac{GM_b}{d_{ab}^3} (I_a \hat{D}_{ab}) \times \hat{D}_{ab}. \quad (72)$$

The body-frame version is

$$N_{a,\text{body}} = Q_a^\top N_{a,\text{in}}. \quad (73)$$

In the implementation described here, (72)–(73) are useful diagnostics of the orientational tendency of the bulge, but they are not added as a separate forcing term in (68).

### H. Orbital dynamics

The pair interaction energy truncated at monopole plus quadrupole order is

$$U_{ab} = -\frac{GM_a M_b}{d_{ab}} - \frac{3G}{2d_{ab}^3} \left( M_b \hat{D}_{ab}^\top I_a \hat{D}_{ab} + M_a \hat{D}_{ab}^\top I_b \hat{D}_{ab} \right). \quad (74)$$

This retains the monopole and monopole–quadrupole couplings, but neglects quadrupole–quadrupole interactions of order  $O(d_{ab}^{-5})$  and all higher multipolar contributions.

The orbital equation is

$$M_a \ddot{X}_a = -\nabla_{X_a} \sum_{b < c} U_{bc}. \quad (75)$$

Equations (1), (66), (67), (68), and (75) form a closed quadrupole tide model.

### I. First-order implementation form

Using the auxiliary variable  $W_a = D_t S_a$ , the quadrupole subsystem may be written in first-order form as

$$D_t S_a = W_a, \quad (76)$$

$$D_t W_a = -2\gamma_a W_a - \omega_{a2}^2 S_a - \sum_{n=1}^{N_{r,a}} c_{an} (S_a - Z_{an}) - \lambda_a E_{a,\text{body}} - \frac{1}{2} \tilde{c}_{J,a} \text{STF}(\Omega_a \Omega_a^\top), \quad (77)$$

together with (67). This form is convenient for standard ODE solvers. In particular, if one implements the inertial quadrupole as  $I_a = c_{Q,a} Q_a S_a Q_a^\top$  in the orbital force and pair potential, then the same model must use the reciprocal coefficient  $\lambda_a = c_{Q,a}/2$  in (66) and (77); otherwise the bookkeeping of conservative energy is inconsistent. The associated torque formula (72) should be treated as diagnostic in this leading-order closure rather than inserted as an additional right-hand-side term in the spin ODE, because the corotational coupling already represented by (70) accounts for the same conservative orientational interaction. Likewise, the same coefficient  $\tilde{c}_{J,a}$  must appear both in the inertia tensor (52) and in the rotational forcing term of (66) and (77). In that reduced closure the coupling is already stored in  $\frac{1}{2} \Omega_a^\top J_a(S_a) \Omega_a$ , so adding an extra spin–shape energy term would double count it and produce a secular drift in “mechanical plus dissipated” energy plots.

### J. Conservative quadrupole energy

In the nondissipative limit  $\gamma_a = 0$  and with all relaxation branches removed, the natural mechanical energy of the quadrupole closure is

$$E_{\text{mech}} = \sum_a \left[ \frac{1}{2} M_a |\dot{X}_a|^2 + \frac{1}{2} \Omega_a^\top J_a(S_a) \Omega_a + \frac{1}{2} \|W_a\|^2 + \frac{1}{2} \omega_{a2}^2 \|S_a\|^2 \right] + \sum_{a < b} U_{ab}, \quad (78)$$

up to the same truncation order used elsewhere in the reduced model and with any chosen normalization of the STF norm. The rotational-flattening coupling is already contained in  $\frac{1}{2} \Omega_a^\top J_a(S_a) \Omega_a$  through the shape dependence of  $J_a(S_a)$ . Therefore the reduced conservative diagnostic must not add a second term proportional to  $S_a : \text{STF}(\Omega_a \Omega_a^\top)$  unless the reduced spin equation is simultaneously upgraded to evolve the corresponding extra contribution to the body-frame angular momentum.

## VII. WHAT THE QUADRUPOLE MODEL CAPTURES

The quadrupole model is the smallest version that still contains the main tidal phenomena.

### A. Tidal bulge

In the slow-forcing limit,

$$S_a \approx -\omega_{a2}^{-2} \left[ \lambda_a E_{a,\text{body}} + \frac{1}{2} \tilde{c}_{J,a} \text{STF}(\Omega_a \Omega_a^\top) \right] \quad (79)$$

up to rheological corrections. The first term is the ordinary tidal bulge, and the second is rotational oblateness.

### B. Lagged bulge

The presence of  $\gamma_a$  and the relaxation branches  $Z_{an}$  implies that the quadrupole is not exactly aligned with the instantaneous forcing. The resulting misalignment drives secular exchange between orbit and spin. In the present closure the diagnostic torque (72) is generally nonzero, while the spin responds through the coupled  $S_a$ -dependence of  $J_a(S_a)$  and the corotational transport terms.

### C. Spin locking and obliquity damping

If the body-frame forcing changes in time, then the bulge is driven around in the rotating frame. In the present leading-order closure, synchronization and obliquity damping arise from the coupled shape dynamics and the  $S_a$ -dependence of the spin inertia, while (72) remains a diagnostic measure of the corresponding orientational tendency rather than a second explicit right-hand-side term.

### D. Circularization

Because the quadrupole is repeatedly driven and dissipated on an eccentric orbit, orbital energy is removed while angular momentum is redistributed between orbit and spin. This yields eccentricity damping and relaxation toward more circular states.

## VIII. RELATION TO REAL PROBLEMS

### A. What is physically meaningful

Despite its simplicity, the model retains several genuinely important features:

- full three-dimensional orbital motion,
- explicit tidal multipoles,
- backreaction of those multipoles on the gravitational field,
- spin–orbit coupling,

- rotational flattening,
- deformation-dependent spin inertia,
- dissipative lag with adjustable frequency dependence.

These features are sufficient to reproduce the leading qualitative structure of tidal evolution.

### B. What remains toy-like

The model is still a reduced model, and this should be stated clearly.

- **No full hydrodynamics.** The interior is not evolved by compressible Navier–Stokes or by a detailed anelastic or viscoelastic continuum model.
- **No internal stratification.** Real planets may contain cores, mantles, oceans, atmospheres, layered viscosities, and phase transitions. These are not represented explicitly.
- **Effective rather than microscopic rheology.** The coefficients  $\gamma_a$ ,  $c_{an}$ , and  $\tau_{an}$  are reduced parameters. They represent an effective linear response, not a detailed microphysical material law.
- **No exact higher-order angular-momentum closure.** The leading quadrupole implementation keeps the deformation dependence of the spin inertia, but does not explicitly evolve the angular momentum carried by the internal deformation. Exact global angular-momentum conservation is therefore not enforced beyond the retained leading order.
- **Small-deformation regime.** The quadrupole model, and the linearized inertia (52), are intended for weak to moderate deformations. Very strong tides would require higher modes and nonlinear corrections.
- **No thermal evolution.** Mechanical dissipation is included, but heating, melting, convection, and feedback on material parameters are not.
- **No permanent triaxial figure unless added.** The model describes fluid-like tidal and rotational bulges. A frozen-in quadrupole can be added, but is not present by default.

### C. Comparison with common tide models

A single damping term proportional to  $D_t S_a$  resembles a constant time-lag description. By contrast, the branch variables  $Z_{an}$  allow a frequency-dependent lag more like a generalized Maxwell or anelastic response. This is still a reduced rheology, but it is more flexible and more realistic than a single constant-lag parameter.

### D. Relation to Love number and quality factor

Planetary tidal dissipation is often parameterized by the quadrupolar Love number  $k_2$  and the quality factor  $Q$ . In the present reduced model, the more fundamental objects are the linear response coefficients appearing in (66). For a harmonic forcing at angular frequency  $\sigma$  in the body frame, one may write the complex quadrupole response schematically as

$$\tilde{S}_a(\sigma) = -\chi_a(\sigma)\tilde{F}_a(\sigma), \quad (80)$$

where  $\tilde{F}_a$  denotes the STF forcing from the tidal tensor and the rotational term, and where

$$\chi_a(\sigma) = \left[ -\sigma^2 + 2i\gamma_a\sigma + \omega_{a2}^2 + \sum_{n=1}^{N_{r,a}} c_{an} \left( 1 - \frac{1}{1 + i\sigma\tau_{an}} \right) \right]^{-1} \quad (81)$$

in the linearized frequency-domain approximation. The static compliance  $\chi_a(0)$  sets the scale of the equilibrium quadrupole and therefore of an effective static  $k_2$ , while the ratio of imaginary to real parts of  $\chi_a(\sigma)$  determines the phase lag and hence an effective  $Q^{-1}(\sigma)$ .

Thus the present parameters  $(\gamma_a, \omega_{a2}, c_{an}, \tau_{an})$  should be viewed as a way to fit or calibrate the complex quadrupolar response of a body over the frequency range of interest, rather than as a unique microscopic material model. With one branch the model resembles a simple constant-lag-like response over a limited band, whereas several branches can be chosen to better match a desired frequency-dependent  $k_2(\sigma)$  and  $Q(\sigma)$ .

## IX. COMPUTATIONAL STRUCTURE

The upgraded quadrupole model remains inexpensive compared with full hydrodynamics.

For  $N$  bodies, the dominant cost of each right-hand-side evaluation remains the pairwise gravitational and tidal interaction, which scales as

$$O(N^2). \quad (82)$$

The additional costs from the refinements are lower order:

- building and inverting the  $3 \times 3$  inertia tensor  $J_a(S_a)$  is  $O(N)$ ,
- the rotational-flattening term is  $O(N)$ ,
- $N_{r,a}$  rheology branches add  $O(NN_{r,a})$  state variables and arithmetic.

Thus the qualitative complexity does not change. The main effect is a modest increase in state size and per-body algebra.

## X. IMPLEMENTATION REMARKS

For numerical work, the quadrupole model is the recommended starting point. A practical state vector for body  $a$  is

$$(X_a, \dot{X}_a, q_a, \Omega_a, S_a, W_a, Z_{a1}, \dots, Z_{aN_{r,a}}), \quad (83)$$

where  $q_a$  is a unit quaternion representing the orientation.

Several implementation details are important:

- The orientation should be advanced with quaternions or a Lie-group integrator, then renormalized as needed.
- The matrices  $S_a$ ,  $W_a$ , and  $Z_{an}$  should be projected back to their symmetric trace-free parts to suppress numerical drift.
- In the conservative limit, the orbital force, pair potential, and any diagnostic torque should be derived from the same truncated potential. In the leading-order quadrupole closure used here, the torque is monitored diagnostically rather than inserted as an additional spin forcing term.
- The shape-dependent inertia  $J_a(S_a)$  should remain positive definite. Since  $J_a(S_a) = J_{a0}I_3 - \tilde{c}_{J,a}S_a$ , this requires  $J_{a0} - \tilde{c}_{J,a}\lambda_i(S_a) > 0$  for every eigenvalue

$\lambda_i(S_a)$ . Equivalently, the deformation amplitude must remain small enough that no eigenvalue of  $\tilde{c}_{J,a}S_a$  reaches  $J_{a0}$ ; approaching that regime signals breakdown of the linearized weak-deformation model.

#### A. Comments on a practical matrix implementation

A straightforward numerical realization stores each STF quadrupole variable  $S_a$ ,  $W_a$ , and  $Z_{an}$  as a full  $3 \times 3$  matrix rather than in a minimal five-component basis. This is slightly redundant, but it keeps the ODEs close to their analytical form: the corotational terms remain simple commutators, and the rotational forcing is just  $\text{STF}(\Omega_a \Omega_a^\top)$ . Numerical drift can then be controlled by re-projecting these matrices onto their symmetric trace-free parts after right-hand-side evaluations or at output times.

The orientation may be represented by unit quaternions  $q_a$  mapping body-frame vectors into inertial-frame vectors. For moderate  $N$ , the orbital force, body-frame tidal tensor, and diagnostic quadrupole torque can all be assembled by vectorized pairwise operations, preserving the  $O(N^2)$  cost of the reduced model. For the particular leading-order closure used here, it is also convenient in the spin update to solve the  $3 \times 3$  linear system

$$J_a(S_a)\dot{\Omega}_a = -\dot{J}_a(S_a)\Omega_a - \Omega_a \times (J_a(S_a)\Omega_a) \quad (84)$$

directly, rather than forming  $J_a(S_a)^{-1}$  explicitly.

For generalized rheology, initializing each branch near  $Z_{an} = S_a$  avoids an artificial start-up transient. The present implementation also follows the leading-order closure adopted in this paper: it keeps the deformation dependence of  $J_a(S_a)$  but omits the separate angular momentum carried by the deformation itself. Consequently, the conservative diagnostic should be constructed from the same truncated reduced energy used by the ODEs, with the same coefficient  $\tilde{c}_{J,a}$  appearing in both  $J_a(S_a)$  and the rotational forcing term. No additional  $S_a : \text{STF}(\Omega_a \Omega_a^\top)$  storage term should be added on top of  $\frac{1}{2}\Omega_a^\top J_a(S_a)\Omega_a$  in this reduced closure. After that correction, remaining small drift in practical integrations is primarily numerical and reflects the use of finite tolerances, post-processed dissipation integrals, and the leading-order angular-momentum closure rather than an omitted conservative energy reservoir. In production runs one should also monitor the smallest eigenvalue of  $J_a(S_a)$  and reject parameter choices or time steps that drive the inertia tensor close to singularity.

### XI. ILLUSTRATIVE DOUBLE-PLANET EVOLUTION

To illustrate the behavior of the quadrupole model, Fig. 1 shows a representative integration for a close double-planet system. The figure is not intended as a parameter survey, but as a compact diagnostic of how the reduced model moves energy and angular momentum between orbit, spin, and tidal shape variables during long-term evolution.

The upper-left panel shows the energy bookkeeping. The dashed green curve gives the instantaneous mechanical energy of the reduced system, while the black curve shows the cumulative dissipated energy. Their sum, plotted in red, remains nearly constant throughout the run, indicating that the

numerical evolution is consistent with the intended dissipative energy balance: mechanical energy is steadily removed from the orbit–spin degrees of freedom and transferred into heat.

The upper-right panel tracks the projection of each planet’s spin axis into the orbital plane through  $|\hat{s} - (\hat{s} \cdot \hat{h})\hat{h}|$ , where  $\hat{h}$  is the orbital angular-momentum direction. After a brief initial transient, both curves remain close to unity, so in this example the spins stay almost entirely within the orbital plane rather than aligning with the orbital normal. The plot therefore provides a direct diagnostic of obliquity evolution in the chosen initial configuration.

The lower-left panel shows the secular orbital evolution. The eccentricity (monotonic purple curve) decreases steadily, demonstrating circularization. At the same time, the semi-major axis (gray dashed curve, read on the right axis) also shrinks, showing that dissipation removes orbital energy and drives a slow inspiral. Small superposed oscillations are visible, but the dominant trend is a smooth long-term decay of both quantities.

The lower-right panel compares the spin angular frequencies of the two planets with the orbital mean motion. Planet 1 starts with a strong transient and then relaxes toward a nearly steady super-synchronous rotation, while Planet 2 stays at a lower but also nearly steady spin rate. Meanwhile the orbital frequency (black dashed curve) increases gradually as the orbit contracts. The panel thus illustrates the expected competition between tidal spin evolution and the changing orbital timescale.

Taken together, the four diagnostics show that even the reduced quadrupole closure reproduces the main qualitative signatures of tidal evolution: a mechanically consistent dissipative energy budget, long-term orbital decay, circularization, and coupled spin evolution on the same integration.

### XII. CONCLUSION

A reduced Newtonian model of tidally deformable pseudo-fluid planets has been constructed in two layers. The first is a full modal framework with arbitrary angular modes, body-frame rotation, deformation-dependent spin inertia, a reduced rotational-flattening coupling implemented through the shape-dependent spin inertia, and generalized rheology. The second is a quadrupole implementation model suitable for direct numerical integration.

Relative to the simplest damped-quadrupole tide model, the present version is improved in three important respects: it allows the spin inertia to vary with the deformation, it includes rotational flattening through the same reduced spin–shape coefficient that enters the spin kinetic energy, and it supports frequency-dependent lag through internal relaxation branches. These additions preserve the ODE structure of the model while making its tidal evolution more physically credible.

The model remains a reduced model and not a full geophysical simulator. Nevertheless, it captures the key orbit–spin–shape couplings needed for tidal bulges, lag, synchronization, obliquity damping, and orbital circularization. It is therefore a useful intermediate framework between point-mass celestial mechanics and full interior simulation.



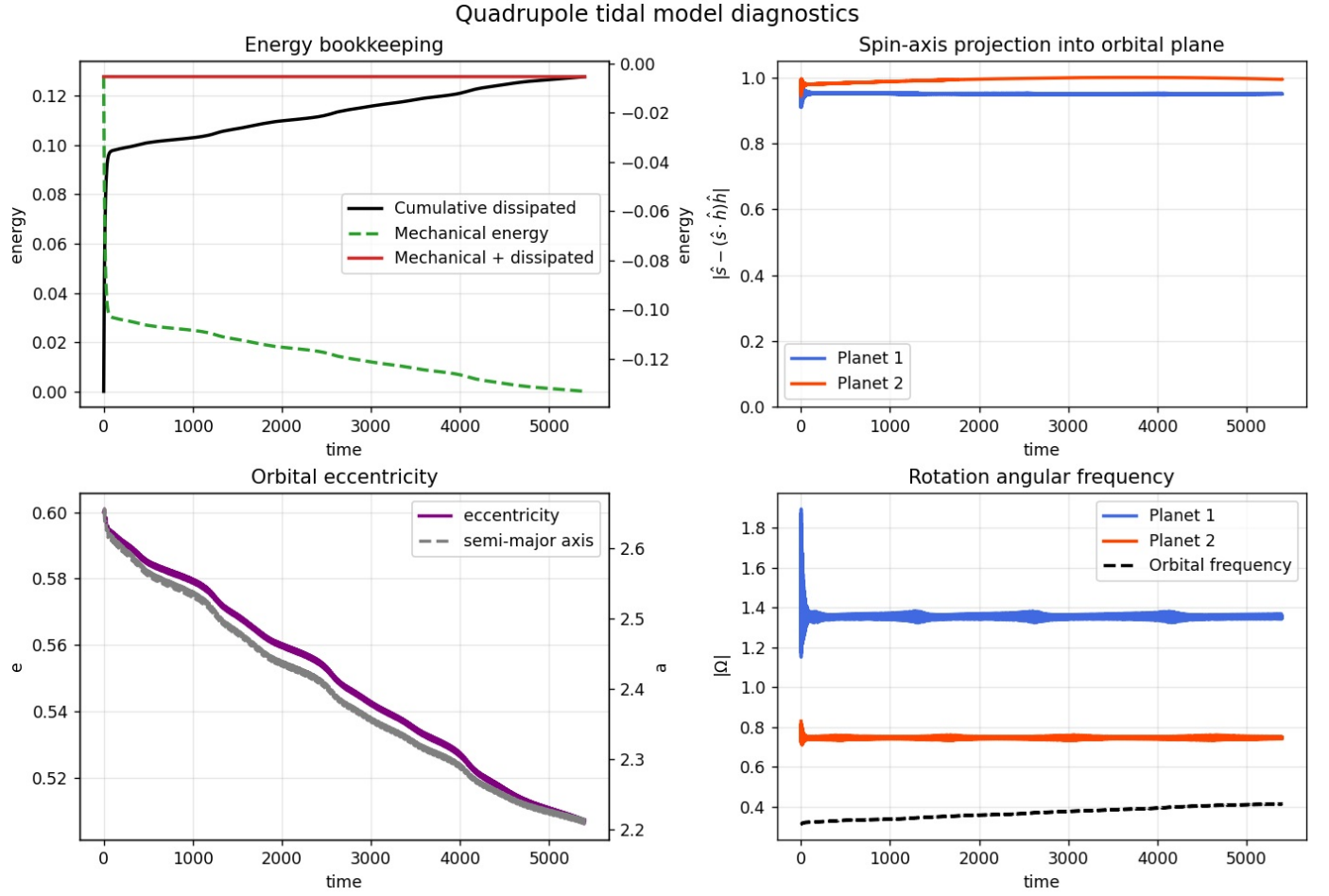


Fig. 1. Example diagnostics for tidal evolution in a close double-planet system evolved with the quadrupole model. The four panels summarize the energy budget, the alignment of the spin axes with the orbital normal, the secular change of orbital eccentricity and semi-major axis, and the evolution of the spin rates relative to the orbital frequency.

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